

Alternative Perception, Broadening Sensing

From Passive to Active: Opening Conversations with Wearables

Mengke Lian

RMIT University

Melbourne, AU

s3805097@student.rmit.edu.au

Abstract

The innate human desire to acquire new abilities dates back to our earliest ancestors [2]. Today, humanity lives in an environment deeply integrated with technology, offering opportunities to extend the boundaries of human capabilities. This integration has led to the emergence of cyborg culture, with "tryborgs"—individuals who attempt to become cyborgs through technology and metaphor—gaining attention[18]. Wearable artifacts can confer individual identity or social status and support the transformation into cyborgs. This research examines whether, by removing the shell of the commodity economy, individuals can create new senses by expanding "extra-corporeal organs," thereby extending perception and expanding boundaries to connect more deeply with their environment. It explores the potential for consciousness and bodily functions to transform into expressions that transcend language and are received universally.

The study takes human sensory ability as a starting point and investigates the effects when individuals imagine an acquired sense through basic technology combined with wearables. This practical research involves speculative and critical design inquiries into reshaping tangible interactions via technology. The core focus includes understanding self-empowerment and constructing new relational methods within community interactions. By shifting the boundaries of data and humanity around perception, the research aims to generate new connections. The proposal seeks to explore the emotional relationships between humans and technology, and among humans themselves, promoting emergent, poetic, and self-generated narratives through open acts of perception.

CCS Concepts

• Applied computing; • Human-centered computing → Interaction devices;

Keywords

wearable technology, sensory design, design research, embodied interaction, affective computing, self-empowerment, auto-ethnography, autobiographical design

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1 Research Background

For the past twenty years, commentators from diverse academic and cultural backgrounds have suggested that the notion of the body and identity as stable and fixed characteristics is gradually fading, especially in Western societies. This perspective is seen in works like Wegenstein's 2006 analysis of the posthuman body and the changing concepts of identity [17]. Wearable technology, as an extension of the body, allows for a broader range of experimentation on the body, enabling quick, easy changes without the need for strenuous physical procedures [1]. Utilizing wearable technology to become a tryborg, akin to prosthetics, could offer a promising, flexible platform adaptable to future changes in both body and mind.

Since the 1960s, body modification practices have emerged in subcultures and movements focused on pushing the boundaries of the human body. These practices have often emerged in subcultures and movements focused on pushing the boundaries of the human body, such as the countercultural movements of the 1960s, the body art movement of the 1970s, and the cyberpunk subculture of the 1980s and 1990s. What stands out across these practices is a clear underlying desire for a body that is more flexible, extensible, and expandable [1]. Recent developments in wearable technology emphasize the materiality of the body and its meaningful relationship with the biological substrate, calling for the inclusion of a sentient body in the future vision of humanity [1].

Building on significant initiatives in 'enhanced' technology, the main objective of this research is to develop, test, and evaluate models for the appropriate use of 'enhanced' technology, focusing on wearable technologies. The research engages with questions of agency and ownership of emergent technologies through practice design research. Although the wearable market has grown intensively in recent years, it is still in an early phase [4]. Customers now seek wearable products that match their identity tags and fulfill beauty and social needs, not just functional enhancements[3][7]

2 Introduction

Despite technological advances, the vision of ubiquitous connected body technologies from science fiction has not materialized. Users are locked in a market-centric loop, with innovation limited by commoditization [12]. Bodies are shaped by bio-political power structures that monitor and extract data [9]. With diverse wearers'



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needs excluded from mainstream conversation, it's time to explore more possibilities.

The proposal links emerging wearable and sensor technologies with embedded augmentation research, drawing on various scientific, religious, and cultural understandings of the senses. For instance, Buddhism describes additional senses beyond the basic five, and scientists are investigating other sensory functions like the magnetic sense [16], immune system-related senses, and interoception [15] [8]. This project focuses on user agency and acquired perception, exploring how humans might enhance their abilities or acquire new sensory organs.

To explore these complex issues, the project seeks answers from potential enhancers themselves, encouraging them to envision acquired abilities through technology. By expanding the range of possibilities for the body and senses, the project questions technological ideologies of progress, power, and control. It aims to ensure that everyone has the right to participate equally, without coercion.

After identifying the needs of potential users of 'augmented' technologies, experiences are prototyped and evaluated through discursive and reflective design practices. This approach explores how technology affects personal, social, and cultural relationships, considering technology users, fanatics, despisers, avoiders, those affected by technology, and those increasingly dependent on it.

3 Problem Statement

Drawing from this research, the proposed project focuses on user agency and acquired perception, bringing practice research to explore the question: how would humans like to "enhance" their abilities or acquire new sensory organs?

To achieve this aim, the research questions are:

1 When the shell of the commodity economy is removed, could design support individuals to create new senses by expanding their 'extracorporeal organs' and by doing so, expand perception and extend boundaries to connect with their surroundings?

1.1 What senses will individuals want to expand? Can wearable devices fulfill the personal desire for "enhancements"?

1.2 How does the development of non-verbal communication and different senses through wearable technology affect personal, social, and cultural relationships?

1.3 How will individuals use 'enhancement' technologies as they become available and empowered? Who will be the potential users of this technology? How will the boundaries between the digital and the humanistic develop?

2 How can design research facilitate the production of new knowledge regarding what diverse individuals want of emergent wearable devices'?

The logic behind these questions revolves around exploring the potential impact and influence of wearable technology in extending human senses, enhancing personal experiences and shaping various aspects of personal, social and cultural relationships. Questions 1.1, 1.2, 1.3 will be subgroups of Q1. The questions aim to delve into the following key themes: 1 is how to extend perception and connect with the surrounding environment, a is a discussion of the individual's desire for augmented and wearable devices, and b is the impact on personal, social and cultural relationships. c is the

utilization of augmented technology and 2 is to generate discussion in the field of design research.

As a way of exploring and understanding the complex questions raised by these high-profile endeavors but crucial to discussing and agreeing on principles for technology development, this project will prototype and evaluate experiences by understanding potential users' needs for "enhanced" technologies and perceptions of personal identity and empowerment, through a reflexive design practice-led approach that prototypes and evaluates experiences to broaden the field and carrier of wearable device applications and explore how this technology will affect personal, social and cultural relationships.

4 Research Methodology

This research embarks on a speculative design and critical inquiry into the reshaping of tangible interactions facilitated by technology, anchored in a transdisciplinary approach that incorporates insights from related fields. The initial section of this chapter situates my study within the realms of autoethnography, practice-based research, and rhizomatic methodology, outlining how these frameworks have sculpted the contours of my investigation.

4.1 Autoethnography

Autoethnography is central to my research methodology. As Ellis describes in *The Ethnographic I: A Methodological Novel About Autoethnography*, the emphasis is placed on "research, writing, story, and method that connect the autobiographical and personal to the cultural, social, and political" [6]. By utilizing "I" along with the dynamics of my research and practice, this approach aids in describing and relating my own personal experiences. In doing so, autoethnography resists the allure of solutionism and the specter of pragmatism to maintain a fusion of research and expression [14].

In the process of empowering oneself through the use of wearable devices, autoethnography also lends authority to the designer. It transforms the individual into someone with something to say and a means to say it, avoiding the need to imagine a specific identity to fit a particular prototype. More than this, within the research, individuals have the opportunity to examine themselves, structure their discourse and practice, while also risking their privileges and perspectives being challenged [14]. This intertwined cycle of construction and deconstruction is not aimed at solving a problem or seeking a definitive answer but rather at exploring a tension and allowing questions to arise within it. This is because in our quest for pure methods and clear answers, we might miss a wealth of knowledge, or we may flatten it, simplifying it into neat, tidy knowledge.

The value of autoethnography lies in its ability to highlight the complexity and nuances of personal experiences, treating them as valid and significant sources of knowledge. In doing so, it challenges the traditional boundaries of academic research, advocating for a more inclusive and diverse understanding of knowledge to accommodate the messiness and unpredictability of lived experiences. This methodology fosters a research environment where ambiguity is not only tolerated but seen as fertile ground for discovery and innovation. It prompts us to question the norms and practices of traditional research paradigms, encouraging a more holistic and

interconnected view of knowledge that respects the complexity of human experience.

In expanding the scope of autoethnography, I also recognize the potential for transformative learning — not just for researchers, but also for the audiences engaging with the research outcomes. By presenting personal narratives that resonate with the broader social, cultural, and political contexts, autoethnography facilitates a process of dialogue through which insights and understandings can be co-constructed. This participatory dialogue enriches the research field, offering avenues for empathy, reflexivity, and the construction of collective meaning.

4.2 Practice-based research

Practice-based research serves as a bridge connecting theoretical inquiry with the tangible processes of creation and experimentation. This methodology emphasizes knowledge production through the act of making, where the design and development of technological artifacts become forms of inquiry in their own right. By engaging directly with materials, crafting tools, and participating in the process, practice-based research allows for a deep dive into subjects, offering insights that might not be accessible through purely theoretical or observational methods. The insights gained from practice are used to understand, inform, and then improve, making it a suitable approach for my iterative practice, which necessitates documenting each iteration rather than focusing solely on the outcomes.

This research employs a practice-based methodology to probe how technology can reshape our tangible interactions and emotional connections with the surrounding world. The iterative cycle of designing, crafting, and testing wearable devices serves not only to explore the potential and limitations of current technologies but also to envision future possibilities for human-computer interaction.

Moreover, practice-based research champions an experiential understanding of technology, positioning the subjective experiences of the designer and users as valuable sources of knowledge. This perspective shifts the focus from viewing technology merely as a tool for efficiency or functionality to considering it as a medium for emotional expression and human experience. Through the development of prototypes and interactive experiences, this research aims to uncover novel ways of perceiving, interacting with, and comprehending our technologically mediated environments.

By emphasizing the value of process over product, this approach not only enriches our understanding of technology's role in society but also opens new avenues for creative exploration and critical reflection on the evolving dynamics of human-technology relations.

4.3 Rhizomatic methodology

Given the multidisciplinary nature of my practice, my prototypes evolve from a cross-pollination of concepts, speculative ideas, and processes, akin to the interconnectedness seen in mycelial networks. One characteristic of the Rhizome is its inherent multiplicity of entryways, exit points, and lines of flight of its own. French philosophers Gilles Deleuze and Félix Guattari used the terms rhizome and rhizomatic as a concept to describe theories and research that have

“no beginning or end; it is always in the middle, between things, interbeing, intermezzo” [5]. A key aspect of Rhizomatic Methodology is its opposition to the ordered hierarchical systems and centralized, homogenized epistemologies produced by arborescent thinking. It strives for a form of effusive breakthrough, seeking to disrupt and proliferate beyond conventional boundaries and structures.

In my research, I employ Rhizomatic Methodology as a metaphor to describe how my ideas iterate, grow, and intertwine as necessary. This occurs as my thoughts weave, sprout, and diverge from each other, but then reconnect as they brush past each other in the iterative structures of their development. From the outset of my study, I attempted to create a model using KUMU, which has continued to evolve and expand. I strive to categorize rather than hierarchize, with similar rhizomes not necessarily needing to be neatly arranged together. Practically, I need to develop multiple experimental components simultaneously, such as code, electronics, and design, with their evolution and final outcomes dependent on the developmental results of each component part. Sometimes, this involves developing multiple prototypes at the same time, which also intertwine with each other.

Herein, I introduce Donald Schön's concept of ‘reflective practice’ as the ability to reflect on action so as to engage in a process of continuous learning [13]. This concept is particularly relevant to my method of working, as my practice-based research involves a continuous learning process that integrates approaches from multiple disciplines, each with its distinct expertise. Within the context of this study, the Rhizomatic Methodology facilitates interdisciplinary engagement across technology, design, and human experience. It acknowledges that reshaping tangible interactions through technology cannot be fully understood from a single-disciplinary perspective. Instead, it necessitates an approach capable of traversing different domains, integrating insights from fields such as design, sociology, and engineering.

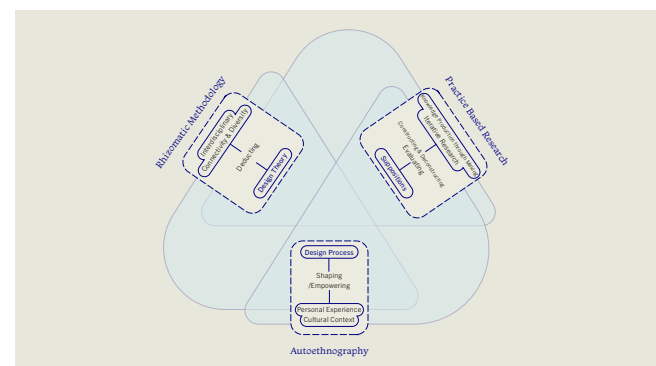


Figure 1: A research methodology model that integrates ethnography, practice-based research, and rhizomatic approaches

5 Practices Conducted and Achievements to Date

My research consists of a series of practice-based projects that delve into the philosophical discourse on the relationship between technology and the body. These projects prioritize perceptual enhancement over commercial applications, echoing William J. Mitchell's approach in *ME++*, which frames research from the perspective of a critically engaged designer rather than a techno-futurist[11]. My work does not aim to plan or predict technology's role in modern society but focuses on reflection, imagination, co-emergence, and invention, questioning existing paradigms.

The practice component is integral to this study, with artifact production and iterative reflection woven throughout. To date, I have conducted six projects, each influencing the others through continuous iteration. These include four wearable devices, one workshop, and one speculative installation design.

Utilizing responsive and emotive wearable technologies, I have developed scenario-based research to engage a broader audience. Human enhancement and sensory exploration are central themes, and through these creative exercises, I seek answers to my research questions. As Howes and Classen note, "The political world has visual, tactile, sensory, and emotional dimensions Modern governance functions in and inhabits the body." Our sensory experiences are never isolated; they shape individual and cultural perceptions[10].

This practice transitions from passive to active engagement, allowing users to reclaim agency over wearable devices. It challenges the predefined sensory 'box' imposed by Western culture, reintroducing multicultural flexibility to sensory interpretation. Potential users should autonomously choose the sensory requirements of their enhanced wearables. Crucially, wearables should not be directly 'controlled' by the wearer but should respond to stimuli, enhancing the agency of surrounding objects and establishing dynamic, symbiotic interactions between user and technology. This approach disrupts the control and domination logic in commercial wearables, cracking the existing design structure as we introduce more-than-human considerations.

By reflecting on these projects, I aim to understand and demonstrate how wearable technology can decentralize design, fostering diverse and subjective perceptions and experiences. This work supports non-linear, interconnected perspectives, contributing to a broader dialogue on technology and human augmentation.

6 Next Steps and Conclusion

This practice-based design research commenced in July 2022 and has now spanned two years. Throughout this period, I have developed numerous prototypes, focusing on creating responsive and emotive wearable technologies. The next phase of this research will involve iterating these prototypes, emphasizing understanding the reasons behind each iteration and establishing clear relationships between different practices. Additionally, further collection and evaluation of user data from workshops will be crucial to refining the designs and enhancing their usability.

The transition from passive to active engagement is a core aspect of this research, empowering users to reclaim agency over wearable devices. This approach challenges the predefined sensory 'box'

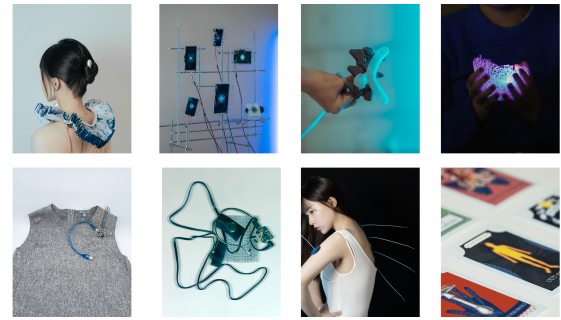


Figure 2: Practices conducted since the beginning of the research

imposed by Western culture, reintroducing multicultural flexibility to sensory interpretation. The goal is to enable potential users to autonomously choose the sensory requirements of their enhanced wearables. Crucially, these wearables should not be directly controlled by the wearer but should respond to external stimuli. This enhances the agency of surrounding objects and establishes dynamic, symbiotic interactions between users and technology. By doing so, this research disrupts the control and domination logic prevalent in commercial wearables, introducing more-than-human considerations and cracking the existing design structure.

Reflecting on these projects aims to understand and demonstrate how wearable technology can decentralize design, fostering diverse and subjective perceptions and experiences. This work supports non-linear, interconnected perspectives, contributing to a broader dialogue on technology and human augmentation. It challenges the notion of monolithic values by promoting diversity and inclusivity in technological design.

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